



Function Point Analysis of SPL Management System

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Strategy

We did not go into deep granular level at first. We had the thought that - we don't really have the perspective that experienced system analysts have to identify function points from requirements. So let's do trial and error.

First, let's try from a broader level and then we will go to a more granular level. But very surprisingly, we got a very reliable result at first shot.

This shows the power of a systematic approach - using this, even a novice can approximate a good design and analysis.

1. About The Project

SPL (Software Project Lab) Management System is a web based application which aims to help the students and SPL board of IIT, DU. This application will help the students to find appropriate supervisors for SPL-1, SPL-2 and SPL-3. Supervisors will be assigned automatically for each SPL through this system according to each SPL's rule. It will help the SPL board to manage SPL's through posting any notice, evaluate continuous marks, and check the progress of students.

1.1 Use Case Diagram of SPL Management System:

Use Case Diagrams

Use Case diagrams give the non-technical view of the overall system.

Level : 0

Name: SPL Management System

Primary Actor: Admin, Teacher(Supervisor, SPL Manager, SPL committee member, Committee head), Student

Secondary Actor: Gmail

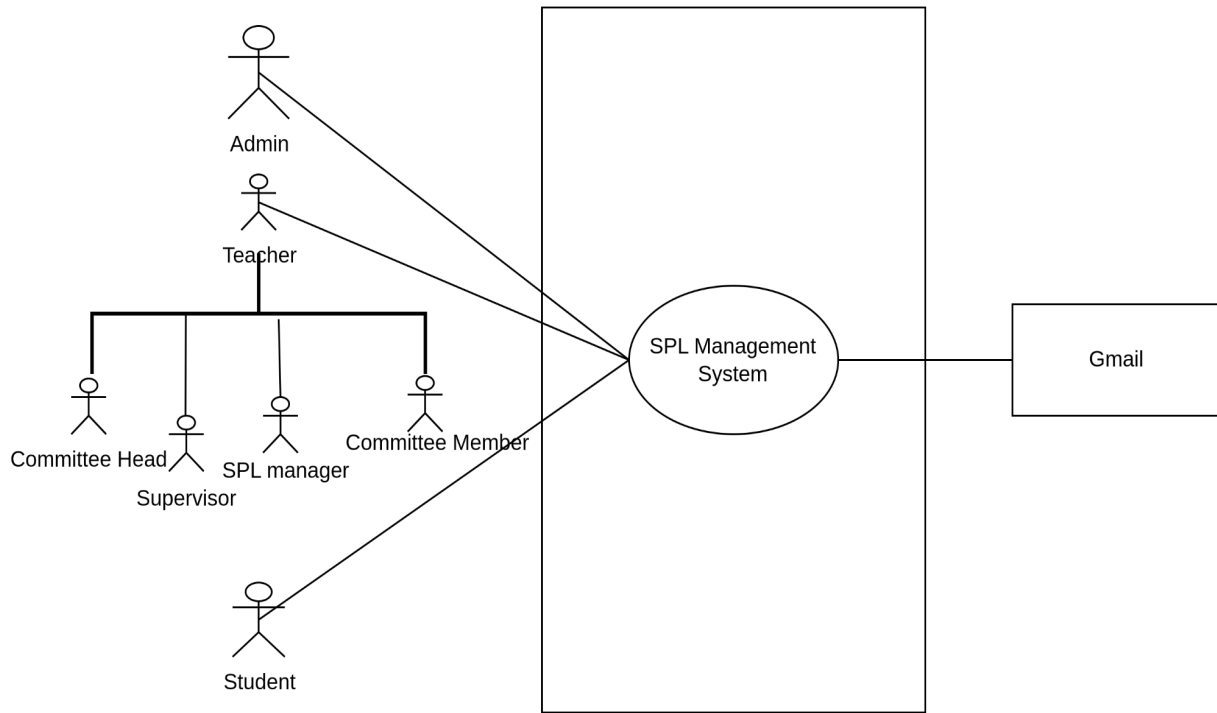


Fig: 1.1 (UCD of Level-0 : SPL Management System)

Level : 1

Name: SPL Management System (Project Overview)

Primary Actor: Admin, Teacher(Supervisor, SPL Manager, SPL committee member, Committee head), Student

Secondary Actor: Gmail

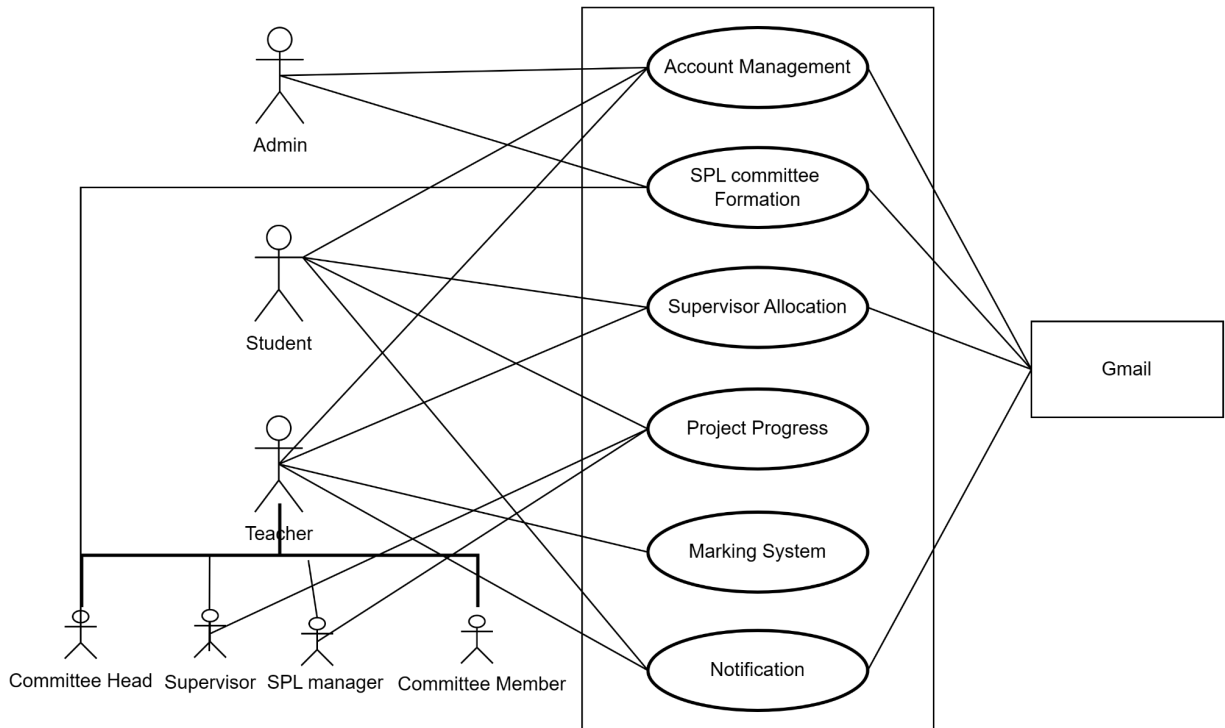


Fig: 1.2 (UCD of Level-1: SPL Management System)

Description:

1. Account Management: Admin will create account for all teachers and a mail will be sent to every teacher with credentials(password). Then teachers will be able to change their password and edit account information after login. Every student has to create their own account(sign up). After creating account they will be able to login and edit their account information.

2. SPL Committee Formation: The admin will create three separate committees for SPL-1, SPL-2 and SPL-3. Every committee will be formed with certain Members, a Committee Head and a SPL Manager.

3. Supervisor Allocation: Every teacher can supervise some students or some groups in these three SPL. There will be three different processes for supervisor allocation of every SPL. Each student will be allocated to 3 separate supervisors across their 3 SPLs.

4. Project Progress: It's basically a task bar. If a student finishes his/her tasks, the supervisor will check those tasks. And on the basis of complete tasks, there will be a progress bar showing the percentage of projects completion.

5. Marking System: There will be some evaluation criteria to evaluate the marks for every SPL. The full evaluation will be in certain marks. The Committee Head will set the individual percentage for every evaluation process.

6. Notification: Supervisor and SPL manager can post any notice. Supervisor's post will be visible to his/her students only. SPL managers can post any notice which visibility will be customized.

Level : 1.1

Name: Account Management

Primary Actor: Admin, Teacher(Supervisor, SPL Manager, SPL committee member, Committee head), Student

Secondary Actor: Gmail

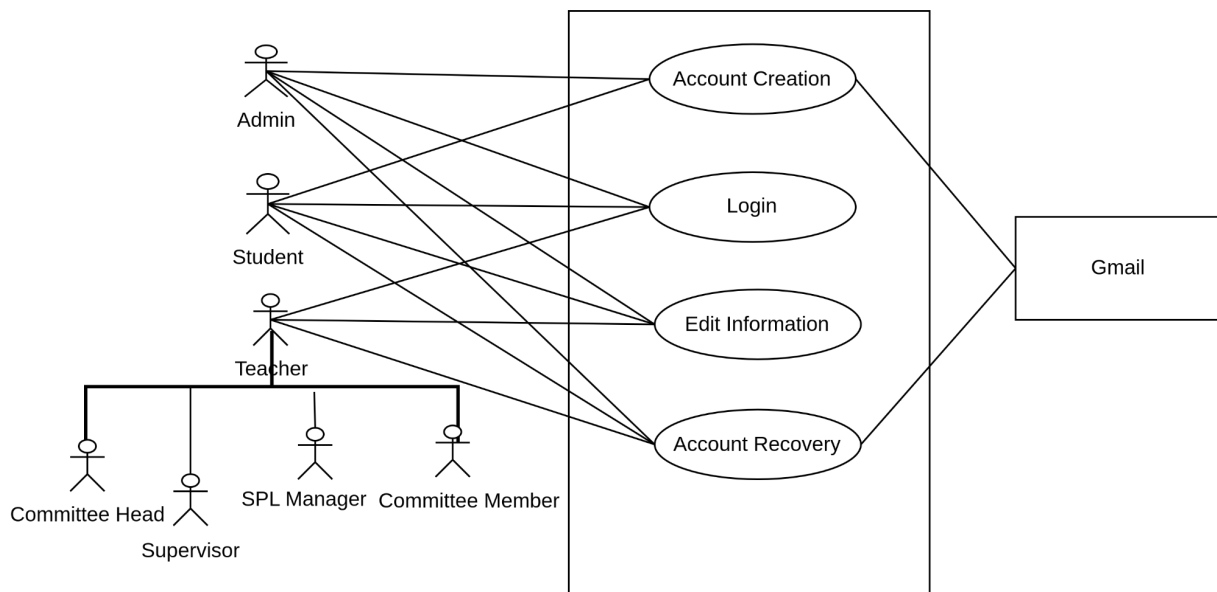


Fig: 1.3 (UCD of Level-1.1 : Account Management)

Description:

1. Account Creation: Admin will create account for all teachers and a mail will be sent to every teacher with credentials(password). Every student has to create their own account with necessary credentials(name, email, password).

2. Login: All users need to be logged into the system. All users can login by providing required information.

3. Edit Information: After logged in every user will be able to edit their account information.

4. Account Recovery: Every user will be able to recover the account if they forget the password.

Level : 1.3

Name: Supervisor Allocation

Primary Actor: Teacher(Supervisor, SPL Manager, SPL committee member, Committee head), Student

Secondary Actor: Gmail

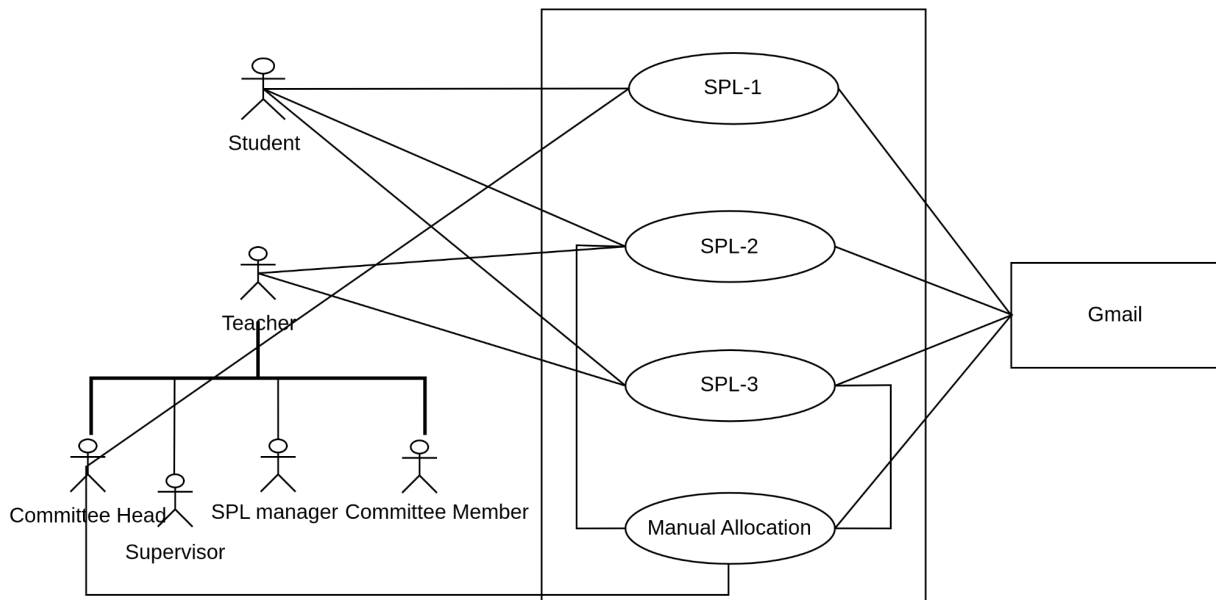


Fig: 1.4 (UCD of Level-1.3 : Supervisor Allocation)

Description:

1. SPL-1: The Committee Head will randomly assign a supervisor for every student.

2. SPL-2: The students will form groups with a specified number of members. Then each group will send requests to teachers to be their supervisor. A teacher can

accept or reject the request. For remaining groups, who didn't find any supervisor, the Committee Head will initiate the supervisor-allocation according to teachers' hierarchy.

3. SPL-3: SPL-3 is individual. Each student will be able to share their interested field/ideas in account details. The system will suggest appropriate supervisor for students by matching their field/ideas with teachers interested field/area. Then each student will send requests to teachers to be their supervisor. A teacher can accept or reject the request.

For the remaining students, who didn't find any supervisor, the Committee Head will initiate the supervisor-allocation according to teachers' field/area.

4. Manual Allocation: The Committee Head will be able to manually allocate supervisor for remaining students of SPL-2 and SPL-3.

Level : 1.3.2

Name: SPL-2

Primary Actor: Teacher(Supervisor, SPL Manager, SPL committee member, Committee head), Student

Secondary Actor: Gmail

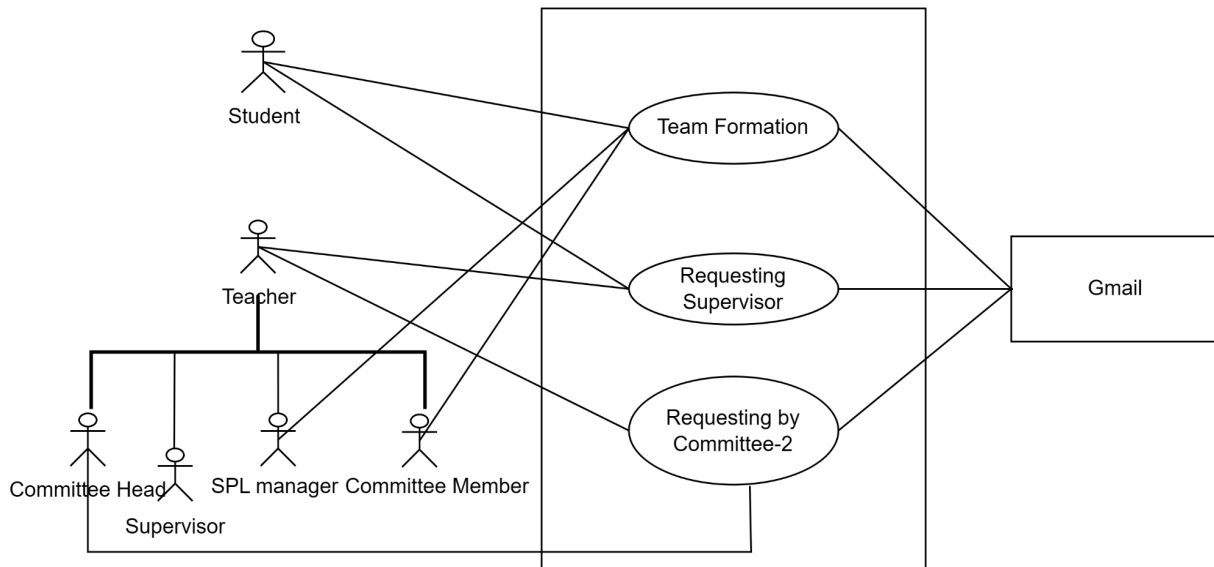


Fig: 1.5 (UCD of Level-1.3.1 : SPL-2)

Description:

- 1. Team Formation:** The students will form groups with a specified number of members.
- 2. Requesting Supervisor:** Then each group will send requests to teachers to be their supervisor. A teacher can accept or reject the request.
- 3. Requesting by Committee-2:** For remaining groups, who didn't find any supervisor, the Committee Head will initiate the supervisor-allocation according to teachers' hierarchy. Teachers will be notified to respond whether he/she is willing to take that group or not. This process can be continued.

Level : 1.3.3

Name: SPL-3

Primary Actor: Teacher(Supervisor, SPL Manager, SPL committee member, Committee head), Student

Secondary Actor: Gmail

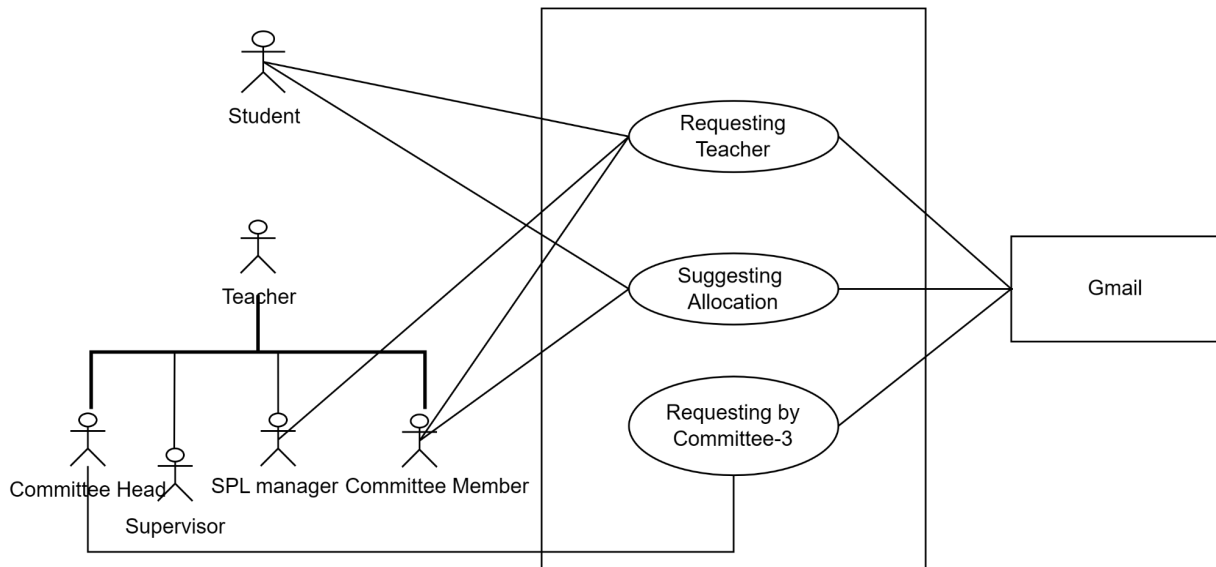


Fig: 1.6 (UCD of Level-1.3.2 : SPL-3)

Description:

1. Requesting Teacher: Then each group will send requests to teachers to be their supervisor. A teacher can accept or reject the request.

2.Suggesting Allocation: The system will suggest appropriate supervisor for students by matching their field/ideas with teachers of the interested field/area.

2. Requesting by committee-3: For remaining students, who didn't find any supervisor, the Committee Head will initiate the supervisor-allocation according to the ideas matched with the teachers idea. Teachers will be notified to respond whether he/she is willing to take that group or not. Teachers have to respond within a given time. This process can be continued.

Level : 1.5

Name: Marking System

Primary Actor: Teacher(Supervisor, SPL Manager, SPL committee member, Committee head)

Secondary Actor: Student

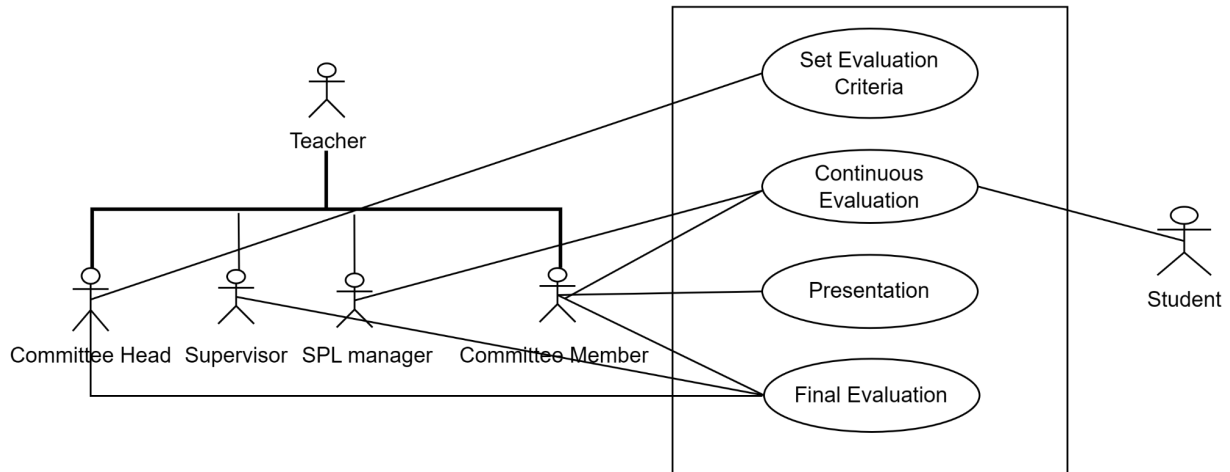


Fig: 1.7 (UCD of Level-1.4 : Marking System)

Description:

- 1. Set Evaluation Criteria:** There will be some evaluation criteria to evaluate the marks for every SPL. The full evaluation will be in certain marks. The Committee Head will set the individual percentage for every evaluation process.
- 2. Continuous Evaluation:** SPL Manager will add the marks of continuous evaluation such as attendance, checking progress.
- 3. Presentation:** There will be several presentations in each SPL. Committee Members will mark individual groups or students.
- 4. Final Evaluation:** After the final presentation all marks will be gathered. According to the authority rule a final grade sheet will be generated.

2. Identifying Function Points

We use the client's perspective to identify functionality here. Hence, they are less granular.

FI D	Functionality of the Software, as described by Client	Type of Functionality
F1	IIT DB (to get all the emails of students and teachers)	External Files
F2	Confirmation Mail (Acc Creation)	External Output
F3	Log In (Password,Mail)	External Input
F4	Teacher List (to make SPL Committee)	Internal Files
F5	Notification(That teacher is in committee)	External Output
F6	SPL-1 Randomization (It will randomly assign supervisor)	External Output
F7	Notify Both Teacher & Students (About Supervisor)	External Output
F8	Show Teacher List (show teachers List in student's side to request them)	External Inquiries
F9	Notify Teacher (about the students & their ideas)	External Output
F10	Requested Student (Teachers will show the students who requested)	External Inquiries
F11	Accept/Reject Students (Response on students requestI) From GUI	External Inputs
F12	All Students List (allocated supervisor,batch name,roll will be shown to check whether any students remaining without supervisor or not)	Internal Files
F13	Admin's Request (request to teachers whether they are willing to take remaining students or not)	External Input
F14	Assigning Supervisor Manually (After admin's Request, If any student remains without supervisor, Admin will assign supervisor Manually)	External Input
F15	SPL-Committee Notice (Notify students about upcoming Event	External Input

FID	Functionality of the Software, as described by Client	Type of Functionality
F1	IIT DB (to get all the emails of students and teachers)	External Files
F2	Confirmation Mail (Acc Creation)	External Output
	or any necessary information about SPL)	
F16	Marks Submission (Supervisor/SPL Manager Marks/Presentation/Final Evaluation)	External Input
F17	Students Marks	Internal Files
F18	Show Student Marks (Show Students marks with individual category)	External Output
F19	Generate Grade Sheet (Students calculated GPA on this Course)	External Output

3. Unadjusted Function Points (UFP)

All features are not equally complex so in this step, we assign their complexity as a categorical estimate and weight them.

3.1 Assigning Complexity Category

FID	Functionality of the Software, as described by Client	Complexity
F1	IIT Database (to get all the emails of students and teachers)	Complex
F2	Confirmation Mail (Acc Creation)	Average
F3	Log In (Password,Mail)	Simple
F4	Teacher List (to make SPL Committee)	Average
F5	Notification(That teacher is in committee)	Simple
F6	SPL-1 Randomization List (It will randomly assign supervisor)	Complex

F7	Notify Both Teacher & Students (About Supervisor)	Simple
F8	Show Teacher List (show teachers List in student's side to request them)	Average
F9	Notify Teacher (about the students & their ideas)	Simple
F10	Requested Student (Teachers will show the students who requested)	Average
F11	Accept/Reject Students (Response on students requestI) From GUI	Average
F12	All Students List (allocated supervisor,batch name,roll will be shown to check whether any students remaining without supervisor or not)	Complex
F13	Admin's Request (request to teachers whether they are willing to take remaining students or not)	Average
F14	Assigning Supervisor Manually & update list(After admin's Request, If any student remains without supervisor,Admin will assign supervisor Manually)	Simple
F15	SPL-Committee Notice (Notify students about upcoming Event or any necessary information about SPL)	Complex
F16	Marks Submission (Supervisor/SPL Manager Marks/Presentation/Final Evaluation)	Complex
F17	Input Students Marks	Simple
F18	Show Student Marks (Show Students marks with individual category)	Simple
F19	Generate Grade Sheet (Students calculated GPA on this Course)	Complex

3.2 Calculating Complexity Score from UFP-C Table

FID	Functionality of the Software, as described by Client	Complexity	Function Type	Weighting Factor	Count
F1	IIT Student & Teachers Database	Complex	Internal File	15	41
F3	Login	Simple		3	
F13	Admin Requests	Average		4	
F14	Edit Final Tabulation	Simple		3	
F15	SPL Committee Notice Submission	Complex		6	
F16	Marks Submission (Supervisor/SPL Manager Marks/Presentation/Final Evaluation)	Complex		6	
F11	Accept/Reject Students Application for Supervision	Average	External Inputs	4	41
F8	Allocated Students List	Average	External Inquiries	4	8
F10	Applicants Requesting Supervision	Average		4	
F2	Email Confirmation	Average		5	
F5	Notification on SPL Committee membership	Simple		4	
F6	Random SPL1 Student Allocation	Complex		7	

F7	Notification on Supervisor Allocation	Simple		4	
F9	Notification on Students Ideas and Additional Info	Simple		4	
F18	Show Student Marks (Show Students marks with individual category)	Simple		4	
F19	Generate Grade Sheet (Students calculated GPA on this Course)	Complex		7	
F4	Setting SPL Committee	Average	External Files	7	22
F12	Final Allocation Tabulation of all students and teachers	Complex		10	
F17	Students Marks	Simple		5	
Total UFP					106

Note that, we got the weighting factor from here [1]:

Function type	Simple	Average	Complex
Internal Logical File	7	10	15
External Interface File	5	7	10
External Input	3	4	6
External Output	4	5	7
External	3	4	6

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4. Degree of Influence(DI)

The following 14 factors influence the implementation aspects of the functionalities, aside from their inheritance of these complexities. These factors are the additional technical requirements that those functionalities must fulfill.

1. Data Communication
2. Distributed Functions
3. Performance
4. Heavily Utilized Hardware
5. High Transaction Rates
6. On-line Data Entry
7. Online Updating
8. End-user Efficiency
9. Complex Computations
10. Reusability
11. Ease of Installations
12. Ease of Operations
13. Portability
14. Maintainability/Facility Change

We will now estimate the influence of these factors by assigning a score between 1-5.

Complexity Influencing Factor	Degree of Influence
Data Communication	5
Distributed Functions	1
Performance	2
Heavily Utilized Hardware	1

High Transaction Rates	3
On-line Data Entry	5
Online Updating	5
End-user Efficiency	4
Complex Computations	2
Reusability	3
Ease of Installations	5
Ease of Operations	4
Portability	4
Maintainability/Facility Change	3
Total DI	49

5. Technical Complexity Factor(TCF)

Technical Complexity Factor (TCF) is a measure of the technical difficulty of a software project. It is used in software estimation to adjust the estimated size of the project based on the technical considerations involved. The TCF is typically calculated by assigning a score between 0 and 5 to each of a set of technical factors, such as the complexity of the system architecture, the use of new or unfamiliar technologies, and the level of integration required with other systems. The scores are then multiplied by weighted values for each factor, and the total of all calculated values is the TCF.

TCF is a key factor in the Use Case Points (UCP) estimation technique. UCP is a size estimation method that is based on the number and complexity of use cases in a system. The TCF is used to adjust the UCP estimate to account for the technical difficulty of the project.

$$TCF = 0.65 + 0.01 \sum_{i=1}^{14} F_i$$

Using this formula, $TCF = 0.65 + 0.01 * 49 = 1.14$

The TCF score of 1.14 indicates that the project is relatively simple. This is because the TCF score is below 1.2, which is the threshold for low complexity projects.

6. Function Points

Function Points (FP) is a unit of measurement for software size. It is used to estimate the cost, effort, and schedule of software development projects. FP is calculated by counting the number of elementary processes in a software system. Elementary processes are the smallest units of functionality that are meaningful to the user.

FP is a widely used software size estimation method, and it is supported by a number of industry standards bodies, including the International Function Point User Group (IFPUG) and the Object Management Group (OMG).

$$FP = UFC \times TCF$$

So Function Points, $FP = 106 * 1.14 = 120.84$

The FP count can be used to estimate the cost, effort, and schedule of the software project by using historical data from previous projects. For example, if the average cost of developing a Function Point is \$100, then the estimated cost to develop a software project with an FP count of 120.84 would be \$12,084.

However, we will use standard COCOMO models for cost estimation due to their reliability.

7. Estimate Lines of Code (LoC) from Function Points

$$LOC = \text{Language Factor} * FP$$

Language	Language Factor	Function Point	Lines of Code
Assembly	320	120.84	38668.8
C	128		14500.8
Cobol	105		12688.2
Fortran	105		12688.2
Pascal	90		10875.6
Ada	70		8458.8
Java	55		6646.2
Python	20		2416.8

We choose Java as the language of choice due to ease of maintainability. So we estimate a total of 6.6KLOC for the software project.

8. Cost Estimation

The COConstructive COSt MOdel was introduced by Dr. Barry Boehm. Software projects under COCOMO basic model strategies are classified into 3 categories, organic, semi-detached, and embedded

A. Organic Mode

A software project is said to be an organic type if-

- Project is small and simple.
- Project team is small with prior experience.
- The problem is well understood and has been solved in the past.
- Requirements of projects are not rigid, such a mode example is payroll processing system.

B. Semi-Detached Mode

A software project is said to be a Semi-Detached type if-

- Project has complexity.
- Project team requires more experience, better guidance, and creativity.
- The project has an intermediate size and has mixed rigid requirements such a mode example is a transaction processing system which has fixed requirements.
- It also includes the elements of organic mode and embedded mode.
- Few such projects are- Database Management System(DBMS), new unknown operating system, difficult inventory management system.

C. Embedded Mode

A software project is said to be an Embedded mode type if-

- A software project has fixed requirements of resources.
- Product is developed within very tight constraints.
- A software project requiring the highest level of complexity, creativity, and experience requirement fall under this category.
- Such mode software requires a larger team size than the other two models.

8.1 Selecting Mode of Software Project

We select **Organic Mode** for the software project. The reasons are:

1. Project team is small.
2. The problem is well-understood and has been solved in the past.
3. Requirements of the project are not rigid. A bureaucratic process can change to accommodate digitization.
4. It is not a Database Management System(DBMS), New unknown operating system or Difficult Inventory Management System.

Mode	a	b	c	d
Organic	2.4	1.05	2.5	0.38
Semi-Detached	3	1.12	2.5	0.35
Embedded	3.6	1.2	2.5	0.32

8.2 Cost Calculation

According to the basic COCOMO estimation model-

Metric	Formula	Calculations
Effort	$a * (KLOC)^b$	$2.4 * (6.6)^{1.05} = 17.4$
Duration	$c * (Effort)^d$	$2.5 * (17.4)^{0.38} = 7.4$
Person Required	Effort / Duration	$17.4 / 7.4 = 2.35$

References

- [1] <https://people.cs.ksu.edu/~padmaja/Project/CostEstimate>
- [2] https://drive.google.com/file/d/1HaAKD1MWnbBX1ZMj_1LDQIbFb-T3n8-E/view